THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET A

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the Sierpinski triangle, the number of small shaded triangles after stage 3 is: 1

(a) 30 **(b)** 27 **(c)** 26 **(d)** 54

2. The number of faces of a cube is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 8

3. The number of faces of a triangular prism is: 1

(a) 6 **(b)** 4 **(c)** 5 **(d)** 7

4. The number of faces of a triangular pyramid (tetrahedron) is: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 3

5. A solid has 6 faces and 8 vertices. Using $F + V - E = 2$, the number of edges is: 1

(a) 14 **(b)** 16 **(c)** 10 **(d)** 12

6. The mean of the data 30, 35, 40, 45, 50, 60, 65 is: 1

(a) 48.4286 **(b)** 44.4286 **(c)** 46 **(d)** 46.4286

7. The mean of the data 100, 120, 140, 160, 180 is: 1

(a) 138 **(b)** 140 **(c)** 142 **(d)** 141

8. The median of 22, 24, 26, 28, 30 is: 1

(a) 29 **(b)** 27 **(c)** 26 **(d)** 25

9. The median of 3, 9, 6, 12, 15, 21, 18 is: 1

(a) 11 **(b)** 15 **(c)** 12 **(d)** 13

10. The mean of 5 numbers is 32. 4 of them are 25, 30, 35, 40. The remaining number is: 1

(a) 28 **(b)** 32 **(c)** 35 **(d)** 30

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is self-similarity? 2

12. In the Koch curve, how does the length of each stage change? 2

13. The mean of 6 numbers is 9. When a seventh is added the mean becomes 10. Find the number. 2

14. Explain 'mean as a balancing point'. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A cube has edge 2 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $4 \times 2 \times 2$. **3**

16. The mean of 5 observations is 32. If 4 of them are 25, 30, 35, 40, find the remaining observation. **3**

17. A cube has edge 3 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $6 \times 3 \times 3$. **3**

OR

A cube has edge 5 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $10 \times 5 \times 5$. **3**

18. The mean of 6 observations is 12. If 5 of them are 6, 9, 12, 15, 18, find the remaining observation. **3**

OR

The mean of 4 observations is 50. If 3 of them are 40, 50, 60, find the remaining observation. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A school logo uses Sierpinski triangles with side 16 cm. **5**

(i) After stage 3, what is the side of the smallest white triangle?

(ii) Count the shaded ones.

(iii) Total shaded area fraction.

OR

(i) A prism has 12 vertices. Name it and find its faces and edges. **5**

(ii) Verify Euler's formula.

20. (i) Find the median of 17 numbers where the sorted 9th value is 42. **5**

(ii) Why?

OR

(i) If each of 9 observations is multiplied by 2, how do the mean and median change? **5**

(ii) Illustrate with 3, 5, 7.

END OF QUESTION PAPER — Total: 40 Marks

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